Research Project – Phase 1: Agent-as-a-Judge on DevAI — Research Gaps and Objective

Name: _____ · Roll No.: _____ · Syllabus unit: Evaluation, safety and governance (LLM judges, cost and latency modelling)

1. Papers read

1. **[Primary]** M. Zhuge, C. Zhao, D. R. Ashley, W. Wang, D. Khizbullin, Y. Xiong, Z. Liu, E. Chang, R. Krishnamoorthi, Y. Tian, Y. Shi, V. Chandra, J. Schmidhuber. "Agent-as-a-Judge: Evaluate Agents with Agents." *Proc. 42nd International Conference on Machine Learning (ICML)*, 2025. arXiv:2410.10934. Code and data: github.com/metauto-ai/agent-as-a-judge; huggingface.co/DEVAI-benchmark.
2. **[Context]** L. Zheng, W.-L. Chiang, Y. Sheng, et al. "Judging LLM-as-a-Judge with MT-Bench and Chatbot Arena." *NeurIPS 2023, Datasets and Benchmarks Track*. (Defines the LLM-as-a-Judge baseline and its known biases.)
3. **[Context]** A. Panickssery, S. R. Bowman, S. Feng. "LLM Evaluators Recognize and Favor Their Own Generations." *NeurIPS 2024*. (Self-preference when the judge and the judged share a model.)

2. Problem and method of the primary paper

Problem. Agentic systems that build software produce a whole workspace (code, files, logs) and a long trajectory of actions, yet they are usually graded only by a final pass/fail signal. Human grading gives step-level feedback but is slow and expensive, and a plain LLM-as-a-Judge sees only a text dump of the output, so it cannot check whether a particular file exists, what it contains, or whether it ran without errors.

Benchmark. The authors build **DevAI**: 55 small AI-development tasks (e.g. "train a CNN-LSTM on RAVDESS and serve it through Flask") with 365 binary *requirements* arranged as a dependency graph (DAG) and 125 softer preferences (§2.2). Three open-source developer agents — MetaGPT, GPT-Pilot and OpenHands, all on gpt-4o-2024-05-13 with a 30-minute limit — are run on every task (§2.3, Table 1). Three expert evaluators label every requirement, first independently and then through a consensus discussion (86.5 person-hours); the consensus is treated as ground truth (§3, Table 2).

Method. Agent-as-a-Judge replaces the single judging call with a small agent built from modules (§4.1): *graph* (map the workspace structure), *locate* (find the files a requirement refers to), *read* (open those files, 33 formats), *retrieve* (pull relevant steps from the trajectory) and *ask* (decide SATISFIED / UNSATISFIED with a justification). Search, planning and memory modules were also tried and dropped (Appendix K). It is evaluated in a black-box setting (workspace only) and a gray-box setting (workspace + trajectory). **Reported results:** black-box alignment with the human consensus is 88.5 / 83.9 / 90.4% (MetaGPT / GPT-Pilot / OpenHands) against 84.2 / 65.3 / 60.4% for LLM-as-a-Judge, comparable to an individual human evaluator (Table 3). The full DevAI evaluation costs $30.58 and 118 minutes against an estimated $1,297.50 for humans (§4.4).

3. Research gaps

G1 — Cost and matched budget (cost/latency). *Evidence: §4.4.* Agent-as-a-Judge costs $30.58 and 118.4 min; LLM-as-a-Judge costs $29.63 but takes only 11.0 min. The two judges therefore spend almost the same money while the agent is about 10× slower, and no tokens or calls per requirement are reported for either. The LLM-as-a-Judge baseline is a single sample; it is never given a matched budget, for example majority voting over several samples or a cheap file-selection step. It is therefore unclear whether the gain comes from agentic evidence gathering or simply from spending more calls on each requirement.

G2 — Single runs and configuration chosen on the test labels (reliability / reproducibility). *Evidence: Tables 3–5, Appendix K.1.* Every alignment number is a single run with no standard deviation, and the decoding temperature and seeds are not given. The module set was chosen greedily — a component was kept or dropped according to its alignment on the OpenHands labels — and the final number is reported on those same 365 labels, so the configuration is tuned on its own test set. Several decisions rest on differences of under two points (90.44% vs 90.16% vs 88.52%, Table 5) that cannot be separated from sampling noise without repeated runs. The text also quotes 92.07% for OpenHands (§4.2), while Table 3 lists 90.16% in that column.

G3 — One model family; judge and judged share it (generality / evaluation). *Evidence: §2.3, §4, released code.* The developer agents run on gpt-4o-2024-05-13, and the paper does not name the judge's backbone in §4 or Table 3; the released code defaults to gpt-4o-2024-08-06. Judge and judged therefore appear to come from the same family, which is exactly the setting in which self-preference has been demonstrated [3]. No second model family and no open-weight judge is tested, so it is unknown whether the method still works when the backbone is a small local model.

G4 — Class-imbalanced labels summarised by raw agreement (evaluation). *Evidence: Table 2, §4.2, Appendix H.* Only 22–45% of requirements are met (Table 2), so a judge that mostly answers "unsatisfied" already scores high: LLM-as-a-Judge gets 84.15% on MetaGPT this way, as the authors themselves note. The headline metric is still raw alignment, with no chance-corrected agreement (Cohen's κ) or per-class F1. The ground-truth labels come from three of the paper's authors, and the benchmark covers one domain with 55 tasks.

4. Objective (addresses G1, with G3 as a secondary effect)

To evaluate Agent-as-a-Judge (locate + read + ask modules, black-box setting) running on an open-weight backbone (Qwen2.5-7B-Instruct, served locally through Ollama) **for** requirement-level judging of the OpenHands outputs **on** DevAI

(55 tasks, 365 requirements, with the released human-consensus labels), **compared against** LLM-as-a-Judge with majority voting over k samples on the same backbone **at a matched token budget**, **measured by** alignment rate and Cohen's κ against the human consensus **and by** LLM calls, prompt + completion tokens and wall-clock latency per requirement.

Feasibility. DevAI, the OpenHands workspaces and trajectories, and all three sets of human labels are public in the authors' repository, so no developer agent has to be re-run. The judge is the only component that calls a model. Workspaces are a few hundred lines of code, so they fit in the context window of a 7B model. The ablation for Phase 2 removes *locate*/*read* (judge sees the same dump as the baseline), which isolates the evidence-gathering component.

Token budget. Agent-as-a-Judge ≈ 3 calls × 365 requirements × 3 runs ≈ 3,300 calls; majority-vote baseline with k set so tokens match (expected $k \approx 3$–5) ≈ 3,300–5,500 calls; ablation ≈ 1,100 calls. Total ≈ 8,000–10,000 local calls, API cost $0 (open-weight model on a local GPU or one Colab session).

Use of language models: I used Claude (Anthropic) to help locate candidate papers, check whether the gaps had already been closed in the 2025–26 literature, and draft this summary; I verified every number against the primary paper and can explain each sentence.